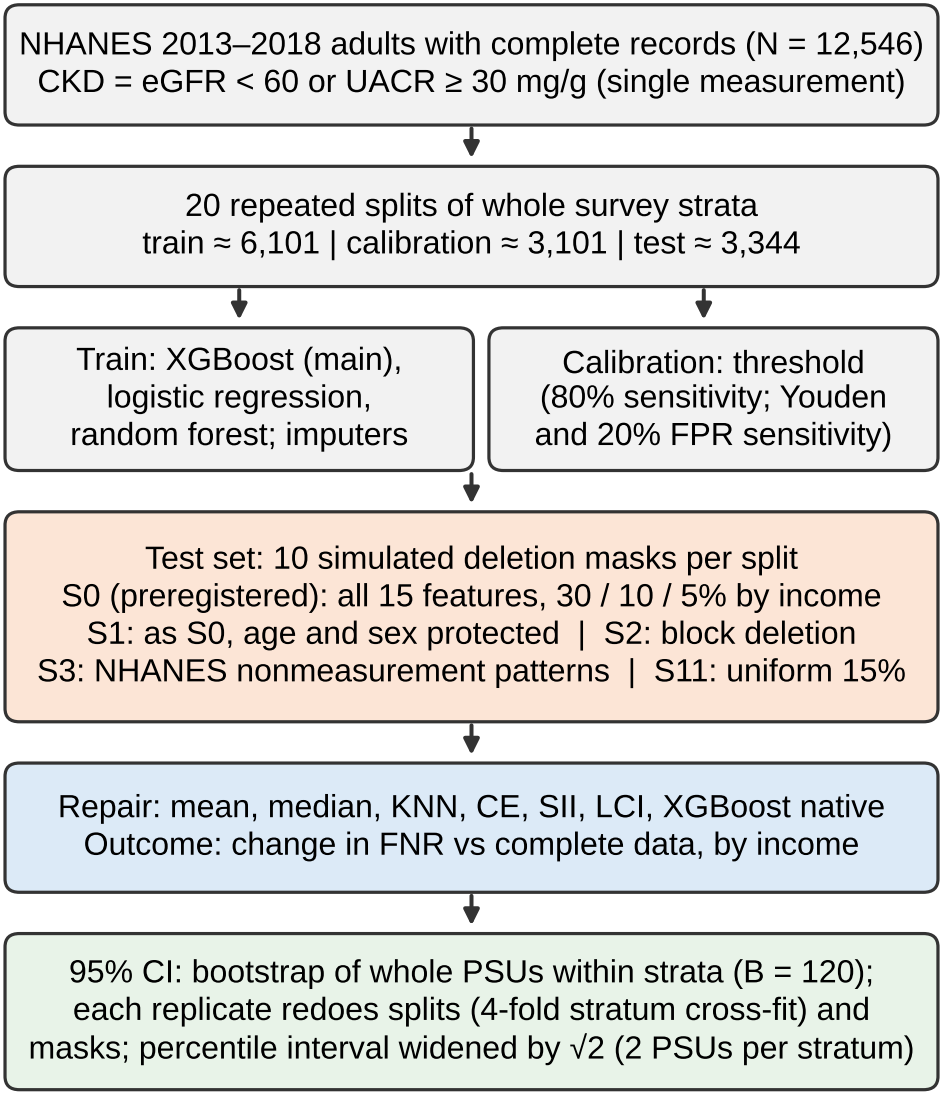
NHANES 2013–2018 adults with complete records (N = 12,546)
CKD = eGFR < 60 or UACR ≥ 30 mg/g (single measurement)
20 repeated splits of whole survey strata
train ≈ 6,101 | calibration ≈ 3,101 | test ≈ 3,344
Train: XGBoost (main), logistic regression, random forest; imputers
Calibration: threshold (80% sensitivity; Youden and 20% FPR sensitivity)
Test set: 10 simulated deletion masks per split
S0 (preregistered): all 15 features, 30 / 10 / 5% by income
S1: as S0, age and sex protected | S2: block deletion
S3: NHANES nonmeasurement patterns | S11: uniform 15%
Repair: mean, median, KNN, CE, SII, LCI, XGBoost native
Outcome: change in FNR vs complete data, by income
95% CI: bootstrap of whole PSUs within strata (B = 120);
each replicate redoes splits (4-fold stratum cross-fit) and
masks; percentile interval widened by √2 (2 PSUs per stratum)